A8 — Imitation Learning & Inverse RL: task, goal, and how it was solved

Why copying an expert's actions and recovering an expert's objective are different things, and what the difference costs. Lab notebook: `experiments/a8_imitation/README.md`.

> **At a glance**
>
> - Behaviour cloning is exactly as good as its data: expert-level on demonstrated states, −70.6 against −8.1 off them.
> - Generalising is not generalising correctly. Feature-based BC answers everywhere and is worse everywhere, drifting off-distribution 24× as often as the expert.
> - IRL followed by RL matches the expert almost everywhere, yet the recovered reward correlates only 0.56 with the true one: the objective is not identifiable, only its optimal policy.

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1. The task

`ladder.md` states three items:

1. Behavioural cloning from a handful of demonstrations; observe compounding error away from the expert's state distribution.
2. A simple inverse RL method (max-entropy IRL) to recover a reward, then run RL against it.
3. Compare the two on states the demonstrations never visited.

2. The goal: the reward was always given, until now

Every phase so far received a reward function. Here it is withheld, and the only signal is a set of expert trajectories. Two things can be done with them, and the gap between them is the phase.

Behavioural cloning is supervised learning: fit $\pi_\theta(a \mid s)$ to the demonstration pairs. It is the obvious approach and it has a specific, well-understood failure.

Inverse RL infers a reward r such that the demonstrations look optimal under it, then plans. The extra step buys generalisation, because a reward plus a model *constructs* behaviour at states never seen, while a cloned policy can only look one up.

Why BC fails away from the data

BC's training and deployment distributions differ, and the difference is caused by BC itself. Training samples come from d^{π^*}, the expert's state distribution; at deployment the agent visits d^{π_θ}, its own. The moment it errs it moves somewhere the expert never was, where it has no training signal, so it errs again.

The standard result (Ross & Bagnell 2010) makes the compounding precise. If the cloned policy errs with probability at most ε under the expert's own distribution,

$$\mathbb{E}_{s\sim d^{\pi^*}}\left[1\{\pi_\theta(s)\neq\pi^*(s)\}\right]\leq\varepsilon \quad\Longrightarrow\quad J(\pi^*)-J(\pi_\theta)\leq O\left(\varepsilon T^2\right)$$

over a horizon T — quadratic, not the $O(\varepsilon T)$ that supervised intuition suggests. The extra factor of T is the distribution shift — an error at step t does not just cost one step's reward, it changes every state that follows. This is why DAgger and friends exist, and why item 1 asks to *observe* the compounding rather than take it on faith.

What MaxEnt IRL actually models

Early IRL matched feature expectations, which is under-determined — many rewards produce the same expectations. Ziebart's maximum-entropy formulation resolves the ambiguity by choosing the distribution over trajectories with highest entropy subject to matching them, which turns out to be

$$P(\tau)\propto\exp\left(\sum_t r_\theta(s_t)\right)$$

Trajectories are exponentially preferred by return rather than strictly optimal, which has a practical consequence worth stating: **suboptimal demonstrations become information rather than contradictions.** Under a hard-optimality model, a single mistake by the expert makes the data inconsistent; here it is merely lower-probability.

The gradient of the log-likelihood is a difference of expected feature counts,

$$\nabla_\theta\mathcal{L}=\boldsymbol{\phi}_{\text{expert}}-\sum_s D_\theta(s)\,\boldsymbol{\phi}(s)$$

where D_θ is the expected state-visitation frequency under the current reward's soft-optimal policy. Computing D_θ needs a forward pass over a discrete state space, which is why this phase uses a gridworld rather than the CartPole the ladder suggests — and why item 3's "states never visited" is an exactly enumerable set here instead of a fuzzy region.

3. The identifiability problem

A reward is not determined by behaviour. Ng, Harada and Russell (1999) showed that potential-based shaping

$$r'(s,a,s')=r(s,a,s')+\gamma\Phi(s')-\Phi(s)$$

leaves the optimal policy unchanged for *any* Φ, because the shaping term telescopes along every trajectory:

$$\sum_{t=0}^{T-1} \gamma^t \left[\gamma \Phi(s_{t+1}) - \Phi(s_t)\right] = \gamma^T \Phi(s_T) - \Phi(s_0)$$

which depends only on the endpoints and so shifts every trajectory's return by the same amount, leaving the ordering — and hence the argmax — untouched. An infinite family of rewards therefore explains any set of demonstrations equally well, and IRL can only ever return a member of that class. Scaling is free too, and a constant offset changes nothing.

This matters beyond the phase. A learned reward model in Part B is an IRL-like object fitted to human preferences, and its *numbers* are routinely read as though they measured something absolute — "the reward went up" — when what is identified is at best an ordering, and only on the distribution the preferences covered. Section 5's measurement puts a figure on the gap.

4. What was done

An 8×8 gridworld with walls and 10% slip, a solvable MDP so every policy can be scored against the exact optimum rather than against the other learner. Six demonstrations from a single start cover 49.5% of free states, leaving 21 never visited.

Three learners, deliberately including a strong form of BC:

- **tabular BC** — counts per state, uniform where unseen. Not a strawman: with no data at a state there is nothing to imitate, and a uniform policy is the honest representation of that.
- **feature-based BC** — logistic regression on coordinates and local wall structure, so that BC is also represented by a version that *does* answer everywhere.
- **MaxEnt IRL then value iteration** on the recovered reward.

Measurements: action agreement with the expert, split by whether a state appeared in the demonstrations; return from demonstrated, all, and never-demonstrated starts; the fraction of a rollout spent off the demonstrated states (item 1's compounding, made numeric); and the correlation between the recovered and true rewards.

5. Results

Tables in the phase README.

BC is exactly as good as its data and no better. Tabular BC agrees with the expert on 100% of demonstrated states and returns -12.3 from the demonstrated start, indistinguishable from the expert's -11.7. From never-demonstrated states it returns -70.6 against the expert's -8.1.

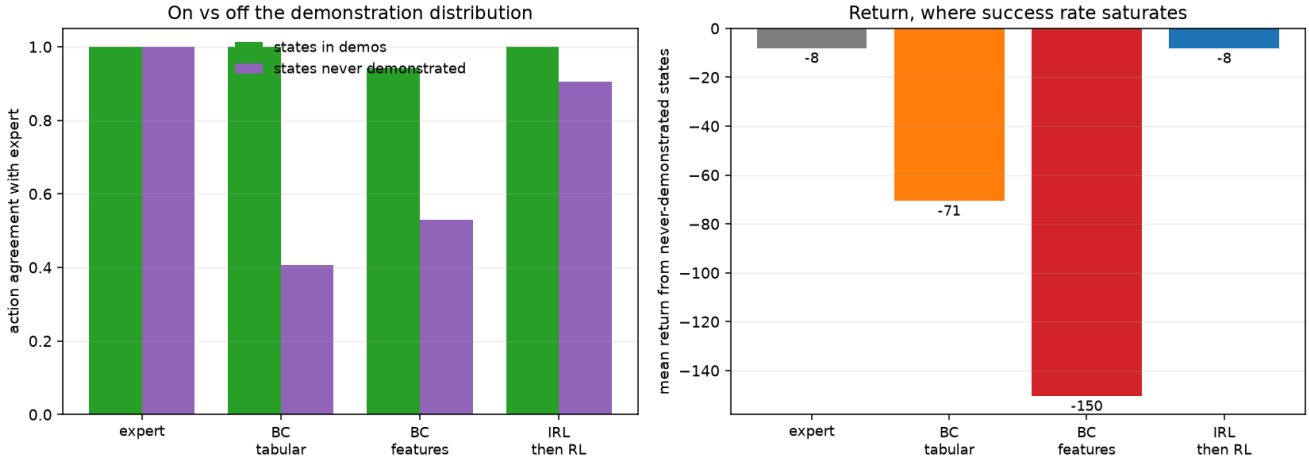


Figure — Agreement with the expert on and off the demonstration distribution (left), and return where success rate saturates and stops discriminating (right).

Generalising is not generalising correctly. The feature-based BC answers everywhere and is worse than the tabular version *at every measurement*, including -70.1 from the demonstrated start where tabular BC is optimal. A smooth function fitted to a policy that depends on local wall structure produces confident wrong answers in place of absent ones, and confident wrong answers steer: its off-distribution fraction is 0.283 against the expert's 0.012. That number is item 1's compounding error made visible — drift into unfamiliar states, then further drift.

Recovering the objective transfers. IRL-then-RL matches the expert essentially everywhere ($-11.5/-4.7/-8.2$ against $-11.7/-4.7/-8.1$) and agrees with it on 90.6% of the 21 never-demonstrated states. It saw those states no more often than BC did. The difference is that a reward and a model can *construct* an action there.

And the recovered reward is not the true one. Correlation $+0.556$, while the policy planned against it is near-optimal — section 3's identifiability result, measured. What IRL returns is a member of an equivalence class: fine for planning, unreliable as a description of what the expert wanted.

A measurement note, the third of its kind in Part A. Success rate reported 0.869 for tabular BC from unseen starts, which reads as respectable; return reported -70.6 against -8.1. Both are correct — a random walk in a small connected gridworld does reach the goal within the step cap, it merely takes forever — but a metric that asks only *whether* the goal was reached cannot see it. Same shape as A3's mean-vs-median and A7's saturated exploitability.

6. What this sets up

- **B2's preference learning** is IRL's cousin: fit a scalar to human choices, then optimise it. The identifiability caveat transfers directly — a reward model identifies an ordering on the distribution its preferences covered, and little else.

- **B4's overoptimisation** is what happens when the equivalence-class member you recovered is optimised far past the region where it was fitted. The gap measured here as a correlation of $+0.556$ is the same gap that opens between proxy and gold metric there.
- **The distribution-shift argument reappears in B5**: offline distillation trains a student on the teacher's distribution and deploys it on its own, which is behavioural cloning's failure with a different name, and on-policy distillation is the DAgger-shaped fix.

7. Where these items come from

item	primary source	why it was picked
1 — behavioural cloning	the ladder's own framing; the compounding-error result is Ross & Bagnell 2010, with DAgger in Ross et al. 2011	It is the simplest thing that could work, and its failure mode is specific, provable and reappears throughout Part B, so it earns its place as a baseline rather than a curiosity.
2 — MaxEnt IRL	Ziebart et al. 2008; the identifiability result is Ng, Harada & Russell 1999	Maximum entropy is the formulation that made IRL well-posed, and its gradient is simple enough to implement from the equations, which is where the assumptions become visible.
3 — compare off-distribution	the ladder's own	It is the only comparison that separates the two methods: on the demonstrated distribution they are both fine, so measuring there would show nothing.

The **lab** pairing (`Practical_RL` week10_planning and yet_another_week) is the ladder's own, which treats this as bonus material beyond the core syllabus.

The **goal statement** in section 2 is quoted from `ladder.md`. The framing in section 3 of a reward model as an IRL-like object inheriting the identifiability problem, and the forward links in section 6, are this repo's rather than claims from the sources.